HATCHSIDE · PROJECT PLAN

HatchSide project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First draft at the kit 1.9.1 upgrade (stage P3), for option A

HatchSide project plan

HatchSide is at stage P3 (TRL 3, proof of concept on paper). The next gate is Amish's decision on the concept choice and on the open calls in HTS-DEC-001; nothing from P4 on starts under the TRL 3 cap (`.kit/PHASE.yaml`). The decision requested: confirm option A, the tripod and hand winch, as the concept (HTS-OPT-001, HTS-DDR-003, proposed, awaiting Amish), and say whether to approach the candidate partners. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and requirements	Design lead	PRB, URS, PRS, RTM	Done for P3; URS is a desk draft pending a co-design partner
STEM check and calculations	Design lead	STM, CAL-001, `sizing.py` , `stem_check.py`	Done for P3; two reviewer follow-ups in HTS-STM-001
Concept options	Design lead	OPT, DDR-003	Done; proposed, awaiting Amish
Detailed CAD of every option	Design lead	`cad/src/model.py` at D3, `cad/fidelity.json` , drawings, STEP, STL	Done for options A and B
Build plan (paper)	Design lead	BLD-001, sheets 101 to 112	Done; a plan, never a build record
Regulatory, feasibility, risk, market	Design lead	REG, FEA, RSK, MKT	First pass done; several rows "to assess"
Partner outreach	Amish	Co-design partner (municipal sewer department, biogas programme)	Not started; none agreed
Prototype build, proof load, corrosion test	Local fabricator, test house	TRL 4 work	Planned only; on hold under the P3 cap

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead time for partner review and revisions	Updated URS, PRS and BOM	5 to 10 working days (estimate)	Not costed (own time)
Parts for one prototype	Built kit at TRL 4	One build	2,838 (BOM estimate; supplier quotes to get)
Fabrication, galvanizing and welding	Made parts	Quote needed	To assess
Proof-load test with a competent person	Proof load to 1.5 times SWL (PR-004)	One day (estimate)	To assess
Coupon test for corrosion	PR-011	Weeks (estimate)	To assess
Legal and standards review	G2, G5, G6, G12 in HTS-REG-001	To assess	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Scaffold	Problem, pitch and first concept	Done (2026-09-30)
P1 Concept	Concept document, first requirements	Done (2026-10-03)
P2 Concept review	TRL 2 decisions recorded	Done (2026-10-03)
P3 Proof of concept on paper	STEM check, concept options, URS and PRS, calculations, D3 CAD of every option, build plan, regulatory, feasibility, risk, market and plan	Done (2026-11-05); concept choice proposed, awaiting Amish
P4 Prototype	Build, proof load, tests	On hold (TRL cap)
P5 Pilot	Field trial with a partner	On hold (TRL cap)
P6 Release	Release gate, public repository	On hold; Amish's decision

Next steps and decision requested

1. Amish: confirm option A or ask for option B to be developed (HTS-DDR-003); decide on HTS-DDR-004's modelling corrections (HTS-DEC-001).
2. Amish: approve approaching the two candidate partners in HTS-PRB-001.
3. When the TRL cap is lifted: buy the winch, rope, chain and shackles with certificates (HTS-DEC-001, items 1 to 5) and write the proof-load test plan; neither is started here.

References

- Sources for the regulatory register: HTS-REG-001, Sources.
- Sources for the market benchmarks: HTS-MKT-001, Sources.

- Sources for the problem: HTS-PRB-001 (NIOSH FACE 89-46; The Wire, 2025; OSHA news release, 2024; Forbes India, 2018; PK Safety; Hach; energypedia; Google Patents US5431248).